

Journey of the Attune NxT Optics

From Prototype to Scalable Product Through Requirements
Driven Architecture



Road Map

Generation 1 problems

Voice of customer

Requirements definition

Architecture decisions

Technical barriers & Engineering solutions

Verification

Outcome

Lessons learned



Gen 1 Problems

Generation 1 demonstrated feasibility but lacked scalability

Field reliability problems - Alignment stability, laser stability

Hard to scale - Long build times, tough to troubleshoot

Difficult to service - Long service procedures to access parts with high failure rates

Difficult to manufacture - Difficult to align to detection



Voice of Customer

Trained and rehearsed how to collect VoC from customers.

Team of 3 people; myself, a scientist, and the product manager visited 8 sites across the US.

Compiled the results and distilled out key Customer Needs

Internal interviews with sales, service, and manufacturing



Key Customer Needs

Customer

More data per cell

Low entry cost but expandable

Service

Quick access to wear parts

Reliable lasers

Sales

Demo with no FSE

Field Upgradeable

Manufacturing

Bring manufacturing in house

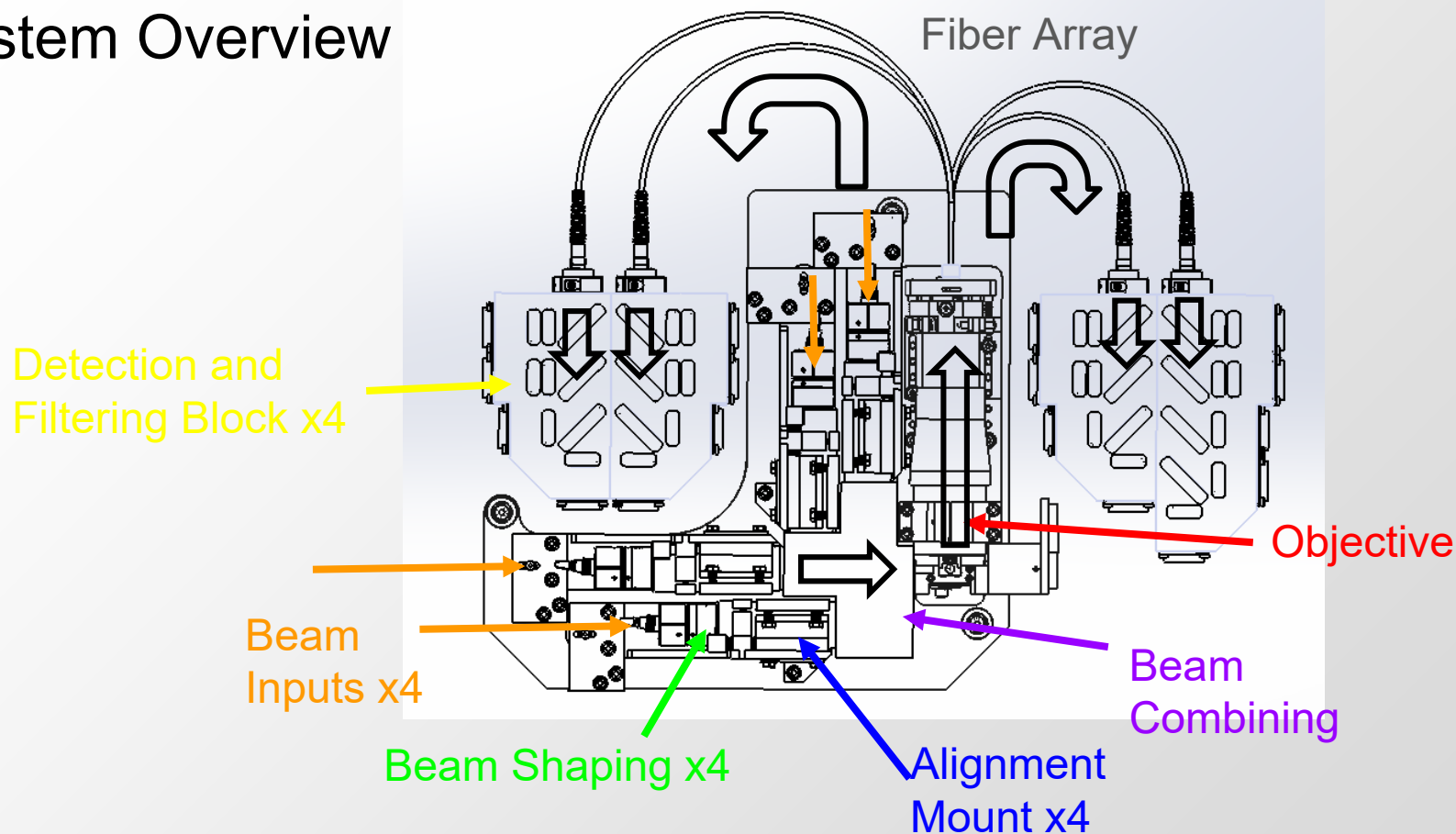
Designed for manufacturability

Translation of Needs to Requirements

Must haves:

- Small footprint -> space claim requirements
- Field upgradable -> time, complexity, and tooling requirements
- Stable alignment -> shock, vibration, and thermal requirements
- Fast build times -> testable subassemblies and ease of alignment requirements
- Easily serviced -> access time to critical parts requirements
- More data -> laser wavelength and detector requirements

System Overview



Feasibility of NUDs

Collection Lens FoV

- 2 to 4 Laser intercepts

Fiber Optic Collection

- Fiber type and size

New Laser Wavelengths

S/N of new detection channels

- Interactions with laser lines and other sources of noise

- New color specific PMTs



Alpha Unit built into (mostly) shell of Gen 1

Architecture Trade Study - Integrated vs Modular

Initial proposal from Coherent was to have a light engine with all lasers integrated and aligned in one box. However, I felt like we needed to evaluate another option where the lasers assemblies were modular by wavelength because of the sales and service requirements.

Key metrics:

- Initial Cost
- Cost of service shelf stock
- Upgrade complexity
- Upgrade cost
- Service complexity
- Instrument complexity



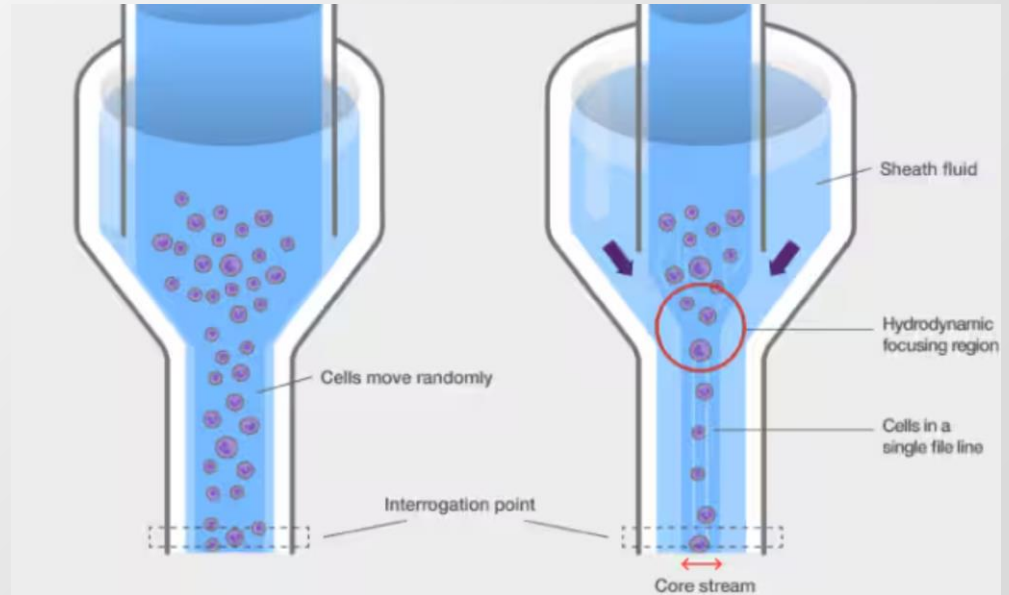
Decision: Modular Laser Assemblies

The modular approach won out due to a number of factors

- Low cost of replacing a single laser
- Much lower cost of only stocking different wavelengths as opposed to the many 2-4 laser flavors of light engines.
- Less system complexity since vertical pitch between lasers was alignable
- Less complexity of supply chain for upgrades since the whole light engine would now be considered “used” as opposed to installing a new laser module
- Minimal increase in initial instrument cost since the upgrades also included laser mounts and beam combining dichroic cubes.

Technical Barriers - Stream Shift

The sample stream position can be influenced by a number of factors and is notoriously unstable within a 10 micron window. To meet MTBF requirements, a better solution than expanding the beams horizontally, and thus reducing power on target, needed to be found.

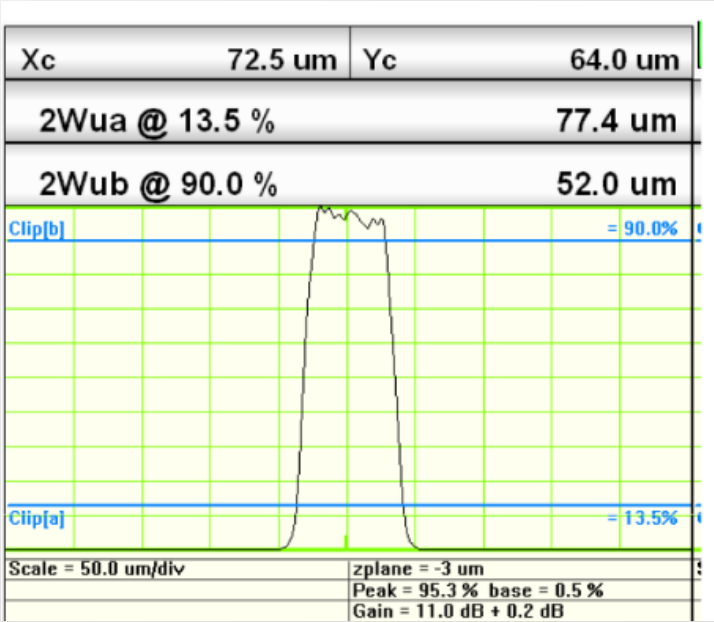


Engineering Solutions - Stream Shift

Implemented flat-top profile with Powell lens

Created tolerance zone for stream shift

Preserved excitation power on target

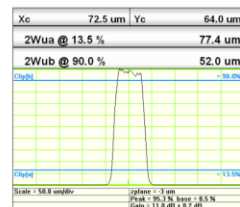


Technical Barriers - Beam Quality

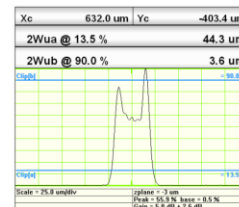
There was an issue with the separation of the vertical and horizontal beam waist focusing at different points for some of the lasers. Not an issue with the light engine, but something I had to figure out for the modular approach

Flat top @ Gaussian Focus

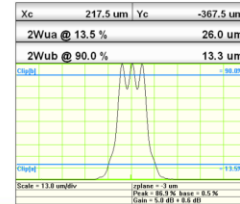
0 Dichroics



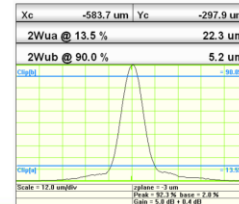
1 Dichroics



2 Dichroics



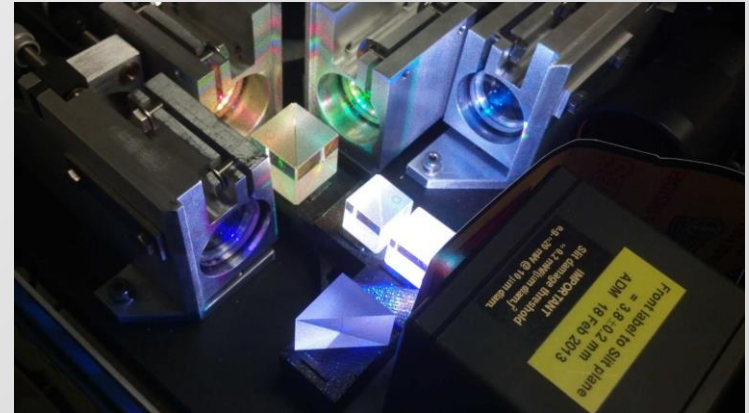
3 Dichroics



Engineering Solutions - Beam Quality

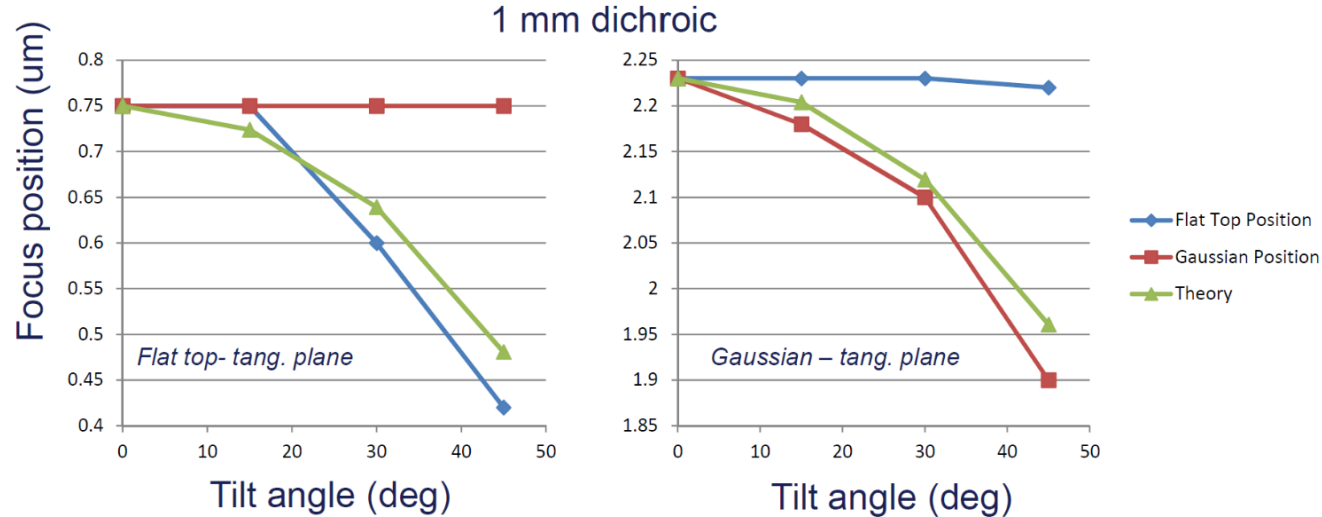
Aberration analysis revealed that 45° dichroic filters used for beam combining induced beam astigmatism and a slight bit of coma proportional to the number of dichroics transited

Patented solution using custom beam combining dichroic cubes solved the issue due to the air-glass interface being orthogonal to the beam propagation.



Engineering Solutions - Beam Quality

Astigmatic Behavior in Orthogonal Planes as Function of Tilt Angle



$$Astig = \frac{t}{\sqrt{n^2 - \sin^2 \theta}} \left[\frac{n^2 \cos^2 \theta}{n^2 - \sin^2 \theta} - 1 \right] \quad \text{Exact}$$

- Dichroic tilt only affects focus in tangential plane (plane where 45deg tilt)

Technical Barriers - Alignment Stability

Gen 1 used adjustable dichroics to steer the beams.

- Unstable over time and temperature for shipped products
- Movable reflective surfaces amplifies any unwanted movements

Engineering Solutions - Alignment Stability

Keys to improving alignment stability:

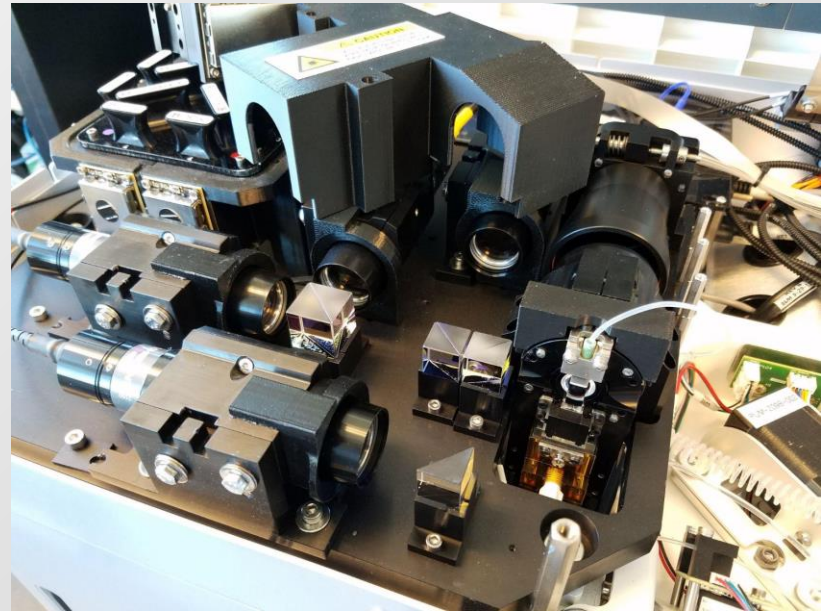
Adjusting the pointing of the laser lens tube instead of moving the dichroic mirrors.

Solidifying the sensitive reflective surfaces

Large contact areas between moving parts of the same annealed material

Spring loading to stabilize movement and minimize misalignment when locking

Micron level adjustment of yaw, pitch, roll, and Z translation achieved using either a 10mm wrench or a flat head screwdriver and a 4mm allen wrench.



Technical Barriers - In-House Manufacturing

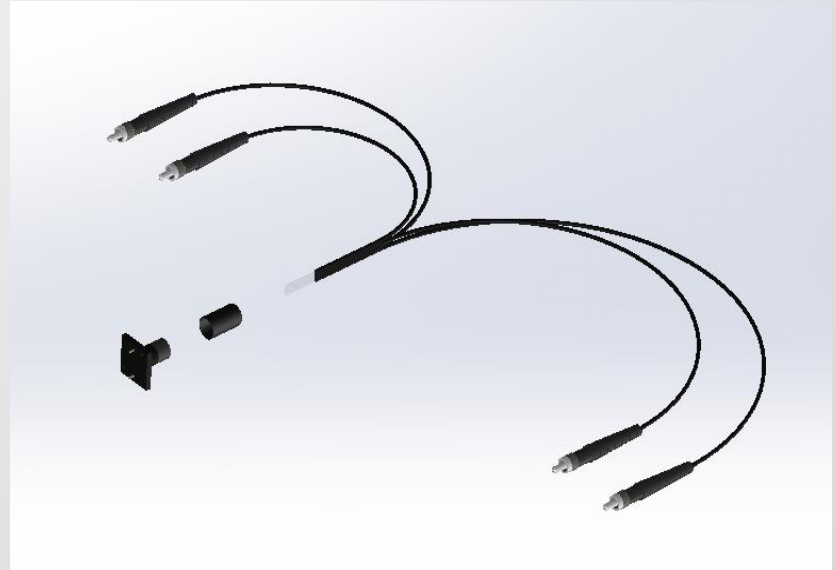
Bringing manufacturing in-house means the optics train needed to be optimized to make the instrument manufacturable in under 20 hours as this is where the contract manufacturer struggled.

- High mag (10x), high NA (1.2) gel-coupled collection lens created a significant tolerance challenge for aligning the flow cell to the lens without clipping optical signal.
- Free space coupling of excitation and emission compounded the difficulty of system alignment.

Engineering Solutions - In House Manufacturing

Decoupling of dependencies creates system margin

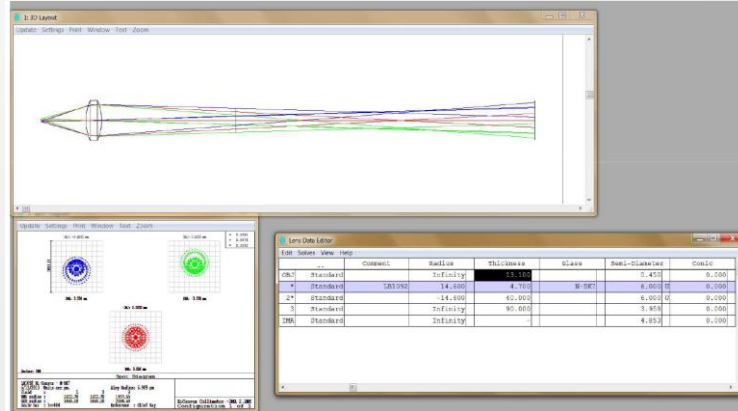
- Fiber optic array to detection blocks eliminated a key alignment dependency
- Blocks could now be positioned as needed as long as the input to the fiber array remained fixed and aligned.



Engineering Solutions - In House Manufacturing

Making appropriate system trade-offs alleviates future pain points

- Careful tolerance analysis of optical, thermal, mechanical, and fluidic variances allowed optimization of downstream optics to reduce need for extremely tight mechanical tolerances between the flow cell and objective



Testing and Verification

Confidence from extensive pre-verification testing

Testing the real use case i.e, Sales demo, Next day Air

Characterization of performance over environmental conditions and beyond

Understanding the tolerances to ensure margin for passing tests

Testable subassemblies key for driving manufacturability of complex systems

Getting early input from all stakeholders (including service, manufacturing, and test) reduces risk of delays and launch complications

Outcome

Scalable platform enabled manufacturing to be stabilized and transferred to Singapore

Install failures reduced from 25% to ~1%

Less than 1 day to do laser upgrade

Business success

- \$10M -> +\$100M in revenue/year
- 2x profit margin on upgrades

KPIs maintained through launch

- Time to Service half of requirement
- Achieved passing stable time to manufacture



<https://youtu.be/AbqiYAh66Mk?si=nd4bmoFVjgeNfHyl>

Lessons Learned

Requirements drive architecture

System level thinking reduces downstream risk

Robustness enables scalability

Trade studies reduce uncertainty

Well written requirements ensure predictable
verification testing

Thank you for your time.

